

Campus Boardwalks

CPU TIME LIMIT

2 seconds

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (2.8)

With all of the construction on campus many of the normal sidewalks between buildings are closed or entirely removed, and students have now taken to cutting directly across fields or around buildings to move between their classes. Since grass grows by the inch but dies by the foot, this has wrecked campus landscaping and caused mud to be tracked into buildings.



After the most recent complaint from the campus janitors, the grounds crew has decided to build boardwalks between buildings to allow students to traverse campus without walking on the grass or mud, but they need help deciding which boardwalks to build. They need to create ways to travel from any building to any other building using only the sidewalks and boardwalks, but want to minimize the time taken to build the boardwalks. This will give them more time to work on their own homework, while also allowing them to fix the landscaping and reduce janitorial work sooner.

They have drafted plans for a number of potential boardwalks while estimating the time needed to build each of them. Each proposed boardwalk will connect two buildings and allow foot traffic in both directions. Can you help them find the minimum number of hours that they need to spend building boardwalks?

Input

The first line of input contains three integers, n , s , and b , respectively the number of buildings on campus, the number of existing sidewalks, and the number of proposed boardwalks. You are given that $2 \leq n \leq 1000$ and $1 \leq s, b \leq 5000$.



Each of the next s lines describes a sidewalk. Each line contains two integers, u_i and v_i , indicating that the i -th sidewalk connects buildings u_i and v_i ($1 \leq u_i, v_i \leq n, u_i \neq v_i$).

Each of the next b lines describes a proposed boardwalk. Each line contains three integers, u_j , v_j , and t_j , indicating that the j -th boardwalk would connect buildings u_j and v_j and would require t_j hours to build ($1 \leq u_j, v_j \leq n, u_j \neq v_j, 1 \leq t_j \leq 1000$).

Output

Output a single integer, the minimum number of hours the grounds crew needs to spend building boardwalks on campus.

If the proposed boardwalk plans do not connect all of the buildings on campus, output **more boardwalks needed!**

Explanation for Sample Input 1

The optimal solution, in this case, requires building the boardwalk from building 5 to building 2 in two hours and from building 6 to building 3 in one hour.

Sample Input 1

```
6 5 4
3 5
1 3
1 5
3 5
2 4
3 4 3
5 2 2
6 1 3
6 3 1
```

Sample Output 1

```
3
```



Sample Input 2

```
3 1 2
1 2
1 2 10
2 1 2
```

Sample Output 2

```
more boardwalks needed!
```